

DATA CLEANING AND INSIGHTS REPORT

Assignment #3 – Data Cleaning and Exploratory Data Analysis

Dataset:

Netflix Movies and TV Shows Dataset

Notebook:

Week_03_EDA_Project.ipynb

1. Introduction

The purpose of this project was to perform data cleaning and exploratory data analysis (EDA) on real-world dataset and extract meaningful insights. The dataset is taken from Kaggle which is Netflix Movies and TV shows. Libraries that we used to perform these analysis are Pandas for data manipulation , Seaborn for styled visualisation , NumPy for numerical operations and Matplotlib for creating charts. The objective is to transform raw data into useful and meaningful data by doing several analysis .

2. Dataset Overview:

Dataset: **Netflix Movies and TV Shows**

This dataset contains information about movies and TV shows available on Netflix

Main attributes included;

- Show id
- Title
- Type
- Director
- Cast
- Country
- Data Added
- Release Year
- Rating
- Duration
- Genre
- Description

3. Data Cleaning Process

Data cleaning is the process of detecting and fixing errors and inconsistency in raw data so analysis give reliable result. There are several steps to clean data and improve the quality of dataset.

Remove Duplicates:

The dataset checked for duplicate values by using `drop_duplicate()` function. No duplicate values found in dataset.

Handle Missing Values:

Several columns have missing values, these are handled by filling text based columns with unknown to avoid losing data. The rating column had special issue as some of duration values present in rating column by default, this issue handled by replacing the invalid values most frequent occurring valid rating in dataset.

Data Type Conversion:

Some data was not in correct format so we convert that data to its correct format like `date_added` column it is originally stored as text so it was converted into proper date using `pd.to_datetime`. Duration column was also not in a useful format, it contained numbers and text together so we split them into two different columns.

Rename Columns Properly:

In this conversion all names were changed to lower case and cleaned by removing extra space.

Outliers Detection:

Outliers in the `release_year` column were found using the IQR method. In IQR method, it finds values that are unusually small or large compared to most of the data. Any title with a release year much lower than the normal range was marked as an outlier. Few titles had very old release year but I kept them because they are real movies that exist in dataset, so I kept it so analysis stay accurate but they notified as outliers.

EDA (Exploratory Data Analysis):

EDA is the process of examining and understanding what the data contains by cleaning it, identifying patterns and trends, finding meaningful insights, and analyzing relationships between variables to better understand the dataset.

These are the several analysis that i perform on pre-processed dataset:

Movies vs TV Shows :

In this analysis we are getting the count of how many movies and tv shows are in this dataset. According to this dataset graph shows Movies count are more than TV shows.

Distribution of Content by Release Year:

In this analysis we get to know how much Netflix content released per year. It shows whether analysis has more older or new content.

Top 10 Content-Producing Countries:

This analysis shows which countries make the most content on Netflix and here we extract the top 10 countries.

Distribution of Content Ratings:

This analysis is based on different content rating. Ratings indicate the recommended audience age group for a movie or TV show.

Top 10 Genres on Netflix:

In this analysis it shows which genre appear the most on Netflix like in this .

Content Added to Netflix Per Year:

It tells the number of titles added to Netflix each year

Top 10 Directors on Netflix:

It counts how many titles each director has on Netflix and shows the top 10.

5. Summary Statistics

- Total records
- Mean values
- Min values
- Max values
- Distribution characteristics

6. Insights :

1. From this dataset we get to know that Netflix's library has approximately 70% of Movies and 30% TV shows.
2. The United States produces the most content around 2500+ titles and then India with around 1000 titles.
3. TV-MA is the most common rating for adult, so TV-MA is the top rating , suggesting most content viewed by adults.
4. The number of titles added to Netflix grew rapidly every year from 2015 and reached its peak in 2019, after that, additions slowed down — most likely because of pandemic.
5. Dramas and International Movies are the top two genre categories on Netflix.
6. No single director has a large share of content. Even the top director has only about 19 titles out of nearly 8,800, which is a very small amount.
7. Between 2018 and 2019, Netflix was expanding aggressively into new countries just before major competitors like Disney plus and Apple TV plus launched in late 2019.
8. While most Netflix titles were released after the year 2000, a small number of titles date back to the 1940s and 1950s, they appear as outlier.

9. A data quality issue was found where a few duration values (e.g., "74 min", "84 min") were incorrectly stored in the rating column and were corrected before analysis.

10. By splitting the duration column into a numeric part and a unit part — minutes for Movies and seasons for TV Shows — it became clear that these two content types cannot be compared using the same measure.

Conclusion:

In this project, I worked on a real-world dataset and gained hands-on experience in data handling and preprocessing. I applied several data cleaning techniques to prepare the data for analysis. By performing Exploratory Data Analysis (EDA), I was able to understand important patterns and trends in the dataset.